SELCO FOUNDATION | CASE STUDY

Solar-Powered Organic Jaggery Making

Mini Jaggery Processing Unit | Pandavapura, Karnataka | 20th May 2026

Demographic Information



Background of the End User

Name	Mrs. Soumya
Age	35 years
Sex	Female
Location	Pandavapura, Mandya District, Karnataka
Education Level	PUC
Family Size	4
End User Typology	Women entrepreneur — organic sugarcane farmer and jaggery processor

Background of the Livelihood

Sector	Food Processing
Type of Livelihood	Jaggery processing entrepreneur
Ownership of Space	Rented workspace
Location of Business	Chikkabyadarahalli, Pandavapura, Mandya Dist.
Land Holding	3 acres
Land Under Cultivation	3 acres — sugarcane (organic)
Primary Output	Organic jaggery; secondary: petty shop

Mrs. Soumya is a 35-year-old PUC-educated organic sugarcane farmer from Pandavapura, Mandya district, Karnataka. She cultivates 3 acres of sugarcane land and initiated the value-added processing of her harvest into organic jaggery approximately six months before SELCO Foundation's intervention. She had taken a workspace at Chikkabyadarahalli for ₹3,000 per month to set up the organic jaggery processing unit and procured the juice crusher and pan for frying.

How SELCO Foundation Identified the End User

SELCO Foundation collaborates with the Puttanaiah Foundation in Pandavapura, and identified the end user and the jaggery community members through this partner network.

The Challenge / Problem Statement

Mrs. Soumya initiated her jaggery processing unit six months before the intervention, but faced compounding barriers that prevented her from operating reliably. Her challenges span energy, infrastructure, finance, and operational safety:

Energy and Infrastructure Barriers

- **Absence of three-phase electrical connection and disputed electricity dues:** The previous occupant of the workspace had left an unpaid electricity bill of ₹30,000. The electricity board removed the meter, leaving Mrs. Soumya in a bind — uncertain whether to pay the legacy dues or invest ₹60,000 in a second-hand diesel engine as an alternative power source.
- **Dependency on diesel:** In the absence of grid access, diesel-powered machines are the conventional fallback — but they carry high recurring fuel and maintenance costs, making the business model unviable at small scale.
- **Erratic power supply:** Even where grid electricity is available, frequent outages disrupt the continuous heating cycle required in jaggery processing, causing product loss and inconsistent quality.

Operational and Safety Challenges

- **Manual drudgery:** Traditional jaggery production relies on continuous manual stirring, sugarcane feeding, and juice handling near open fire — physically demanding and time-consuming work.
- **Poor workspace safety:** Inadequate ventilation, lack of protective barriers near boiling juice, unprotected electrical wiring, and absence of mechanised stirring create significant occupational hazards.
- **Hygiene and quality control:** Unstructured workspace layout and manual handling increase contamination risk and result in inconsistent product quality.

Prior to Intervention — Base Ecosystem Conditions

Ecosystem Component	Pre-Intervention Condition
Household Income (monthly, pre-intervention)	Primarily from sugarcane farming; jaggery unit non-operational due to energy barriers
Energy Source	No electricity connection; the connection was removed due to dues pending from the previous tenant
Hours of Power Cut	Irregular power supply
Cost of Energy	Grid bill disputed; ₹30,000 pending dues, meter removed by electricity board
Finance	Accessed the Government of Karnataka's Secondary Agriculture Scheme (₹2 lakh grant). Personal savings of ₹4 lakh contributed toward machinery.
Training & Capacity Building	Mrs. Soumya had practical knowledge of organic sugarcane cultivation through a champion farmer, Mr. Devagouda, who is an expert in organic farming.
Technology	No functional processing equipment at the time of SELCO's intervention. Conventional approach would require a diesel engine (₹60,000) or three-phase grid connection (₹1–2 lakh) to run a 5 HP crusher — both cost-prohibitive. Traditional jaggery making is fully manual and wood-fire based.

Solution / Intervention

Technical Specifications

SELCO Foundation designed and installed a solar direct-drive system that runs the 5 HP three-phase sugarcane crusher directly from solar panels during daytime hours — completely bypassing the need for a grid connection, diesel engine, or large battery bank. This two-tier energy architecture is the core innovation: the high-power primary load (crusher) runs with solar panels directly via an AC direct-drive system during the 4–5 hours of peak solar generation, while a separate battery-backed auxiliary system powers the boiler blower, juice lifting pump, lighting, and other low-load equipment.

Component	Specification	Role / Notes
Solar PV Array	6.5 kWp	Powers crusher directly via AC direct drive
AC Direct Drive System	For 5 HP, 440 Vac, 3-phase	To run the crusher directly by solar panel
Battery Bank	2.5 kWp, 150 Ah, 4 Nos	Auxiliary loads only (boiler, pump and lights)
PCU / Inverter	5 kVA, 48 V	Manages battery-backed auxiliary system
Sugarcane Crusher	5 HP, 440 Vac, 3-phase; 250–300 kg/hr	Primary juice extracting machine
Boiler / Heating Pan	0.5 HP + 0.5 HP, single-phase (two-stage)	First stage: 3 hrs; second stage: 1.5 hrs
Bagasse	Bagasse used as a fuel	Heating energy is self-sustaining; zero fuel cost
User Access	Panel mounted at accessible height	Weekly cleaning by entrepreneur; low maintenance

Production Process (Batch Mass Balance)

- 500 kg of raw sugarcane is fed into the crusher per batch, yielding sugarcane juice.
- 500 kg of raw sugarcane gives 400 litres of juice: 200 litres is transferred to the first boiler pan and heated for 3 hours; another 200 litres is transferred to a second funnel pan for a further 1.5 hours of heating.
- Conversion ratio: 10 kg raw sugarcane → 1 kg jaggery (10:1).
- Each batch of 200 litres of sugarcane juice yields approximately 25 kg of jaggery.
- Daily capacity: 4–5 hours of solar-powered operation processes approximately 1,000–1,500 kg of sugarcane juice crushing.
- Bagasse (sugarcane fibre residue) is fed back as fuel for the fire chamber manually — the heating process is self-fueling with zero additional fuel cost.



Product Varieties

- 1 kg bucket jaggery
- Half-slab jaggery
- 100 gm jaggery cubes
- 1 kg jaggery powder
- Fresh sugarcane juice (retail: 5–6 glasses/day @ ₹10/glass)

Financial Specifications

Component	Total Cost (₹)	Govt. Subsidy (₹) + Loan	Self-Contribution (₹)	SELCO Foundation (Pilot) (₹)	% of Total
Machinery (Crusher + Boiler/Pan) + BE	8,00,000	2,00,000 + 3,00,000	3,00,000	0	53%
Solar System (6.5 kWp + PCU)	7,20,000	–		7,20,000	47%
Total Investment	**15,20,000**	**5,00,000**	**3,00,000**	**7,20,000**	**100%**

Business Model

Production Economics

Particulars	Unit	Value
Crusher rated capacity	kg/hr	250–300
Daily operating hours (solar)	hrs/day	6–8
Raw sugarcane processed/day	kg/day	500–1,000
Sugarcane juice yield	Litres per tonne	~800 approximately
Jaggery conversion ratio	kg sugarcane → kg jaggery	10:1
Jaggery output per day	kg/day	50–100
Working days per month	days/month	15–20
Jaggery output per month	kg/month	1,000–2,000
Jaggery output per annum	kg/annum	12,000–24,000



Revenue and Cost Breakdown

Item	Unit	Amount (₹)	Notes
Selling Price (wholesale organic jaggery)	per kg	90	2x conventional jaggery price
Revenue (full capacity, per month)	per month	1,80,000	2,000 kg × ₹90
Revenue (per annum)	per annum	21,60,000	Full capacity
Raw Material Cost (sugarcane @ ₹5/kg)	per month	1,00,000	20,000 kg/month @ 20 days
Labour (2 workers × ₹350/day × 20 days)	per month	14,000	₹7,000 per month on record
Packaging	per annum	6,000	Variable cost
Electricity / Fuel	per annum	0	Solar + bagasse; zero recurring energy cost
Workspace Rent	per month	3,000	Fixed cost
Total Expenses (per annum)	per annum	**12,90,000**	All variable + fixed
Operating Profit before EMI (per annum)	per annum	**8,70,000**	
EMI (per annum)	per annum	3,16,694	₹26,391 × 12
Net Profit after EMI (per annum)	per annum	**5,53,306**	~₹46,109/month
Monthly Break-even Volume	kg/month	1,466	Unit breakeven; current capacity ≥ this
Payback Period	years	4.3	Within 5-year ideal threshold

Market Linkages

- Retail / other channels: 70–80 kg per month at ₹90/kg.
- Sugarcane juice retail: 5–6 glasses @ ₹10/glass (supplementary, footfall-based).

Outcomes and Impact

Overall Qualitative Outcomes

Outcome Area	Pre-Intervention	Post-Intervention
Energy Access	No three-phase connection; diesel dependency; unreliable grid.	Fully solar-powered crusher; zero grid dependency for primary load; reliable 4–5 hr daily operation. Saved ₹3,000 electricity bill and ₹4,500 diesel bill.
Income	Unit non-operational; income from raw sugarcane farming only	Net profit ₹5,53,306/year after loan EMI; ₹46,109/month average
Market Access	Could not fulfil B2B orders; limited retail volume ~70–80 kg/month	B2B supply to 3 organic retail chains in Bengaluru; reliable monthly volumes
Finance	No formal loan; ₹30K legacy electricity dues as barrier	₹2 lakh government grant leveraged
Women Empowerment	Aspirant entrepreneur blocked by infrastructure barriers	Active owner-operator; B2B relationships; independent financial decision-maker
Drudgery	Fully manual crushing and processing; physical strain	Mechanised crushing; reduced manual effort; safer working environment
Product Quality	Inconsistent; unable to produce at scale	Standardised organic jaggery in 4 product formats; premium market positioning
Environmental	Potential diesel engine use (50 kg CO_2e/year per 10 L diesel/day)	18 tonnes CO_2e avoided per year; bagasse reuse eliminates external fuel cost

Specific Quantitative Impacts

Indicator	Pre-Intervention	Post-Intervention	Change
Monthly jaggery production	0 kg (unit not operational)	1,000–2,000 kg/month	+1,000–2,000 kg
Monthly revenue	~₹6,300 (farm produce only)	~₹1,80,000	+₹1,73,700
Net monthly profit (after EMI)	–	~₹46,109	New income stream
Electricity bill	₹0 (no meter/connection)	₹0 (solar-powered)	Zero recurring energy cost
Fuel cost	₹0 (unit not running)	₹0 (bagasse reused)	Self-sustaining fuel loop
CO_2e avoided	–	18 tonnes/year	Diesel displacement
Break-even production volume	–	1,466 kg/month	Achievable at 75% capacity
B2B market relationships	0	3 organic retail chains	+3 institutional buyers

Testimonial

> "I was going to buy a diesel engine and, in a confusion, to get the three phase connection – Somehow solar energy intervention by SELCO Foundation rooted me onto a sustainable way to produce the organic jaggery."
>
> **— Mrs. Soumya, Organic Jaggery Entrepreneur, Pandavapura, Karnataka.**

Way Forward

Considering the strong potential of the solar-powered organic jaggery unit, SELCO Foundation will further strengthen the model through operational improvements, safety enhancements, product standardization, and market development support. Focus will be given to improving ventilation, workflow layout, packaging quality, electrical safety, and mechanization to reduce drudgery and improve efficiency.



SELCO Foundation will also explore interventions such as solar drying, branding, premium packaging, and stronger market linkages to enhance product shelf life, value realization, and business sustainability. The successful implementation of this model can serve as a scalable demonstration for promoting women-led, clean-energy-based agro-processing enterprises across other sugarcane-growing regions.